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1. Introduction & Problem Statement

1.1 Project Title

MoodStabilizer: Emotion-Aware Adaptive Bedroom Environment

1.2 Introduction

Daily life places individuals under significant emotional pressure. Teenagers, young adults, and adults often experience loneliness, stress, sadness, anger, or emotional instability without having consistent access to support. Traditional smart homes focus primarily on automation and data collection, but they do not understand the emotional state of the user, nor do they provide therapeutic or stabilizing responses. In environments where people spend most of their time, especially bedrooms, there exists a need for intelligent systems capable of detecting emotional changes and providing meaningful support in real time.

The **MoodStabilizer** project proposes an affective, intelligent bedroom capable of monitoring behavioral signals, inferring the user's current mood, and autonomously adapting the environment to help regulate emotions. By simulating key sensor data and user behaviors, the system aims to demonstrate how IoT and AI can support emotional stability, provide comfort to individuals without continuous human support, and potentially assist mental-health professionals with non-intrusive behavioral insights.

1.3 Problem Statement

Despite major advancements in smart home technologies, existing systems remain reactive and limited. They collect data such as movement or presence, but they do not interpret human emotions nor attempt to provide emotional regulation. Current home assistants and automation platforms respond only to user commands or environmental triggers; they cannot recognize sadness, anger, fatigue, or distress, nor can they intervene at the right moment to improve well-being.

There is currently **no integrated system** that:

- interprets multi-modal behavioral signals to infer mood,
- responds autonomously to stabilize emotional states, and
- continuously learns and adapts to personal habits over time.

This lack of emotional intelligence in home environments leaves a significant gap, especially for individuals who may struggle with emotional regulation, loneliness, or early signs of depressive conditions.

1.4 Scope of the Project

MoodStabilizer will be implemented **entirely in simulation** to ensure accessibility, safety, and feasibility within the academic timeline. The project focuses on **a single bedroom environment** and integrates simulated sensors including:

- motion tracking,
- door-slam intensity levels,
- footsteps cadence and pressure,
- facial emotion embeddings (privacy-preserving),
- smartphone usage patterns (time spent browsing, app opening frequency),
- Other people may enter the room,
- They are treated as “secondary entities”,
- Their data is minimal and non-intrusive,
- System only asks the user for identity if a person appears multiple times.

A mobile agent will be available on the phone as part of the simulated environment, allowing the system to analyze screen-time behavior and communicate with the user when necessary.

No real hardware is required, and all sensing, processing, and responses will be implemented through virtualized data streams and a software-based environment.

The system also accounts for secondary occupants entering the bedroom. Non-primary visitors are tracked only at a minimal behavioural level to detect presence, without inferring their mood. If an entity repeatedly appears and interacts with the user, the system privately asks the user to identify their relation, ensuring personalization while maintaining strict privacy.

1.5 Research Questions

1. Can a combination of behavioral and smartphone-usage signals be used to accurately infer a user's emotional state in a simulated bedroom environment?
2. Can an intelligent agent autonomously select and execute environmental interventions that improve or stabilize the user's mood?
3. Does personalizing the model through continuous learning significantly improve mood-detection accuracy and intervention relevance over time?

1.6 Project Objectives

- **O1: Mood Classification**
Develop a model capable of classifying multiple emotional states—such as sadness, anger, fatigue, happiness, and neutrality—using simulated multi-modal behavioural data.
- **O2: Adaptive Environmental Response**
Design an intelligent decision engine that selects appropriate interventions (e.g., lighting changes, calming prompts, activity suggestions) to stabilize or improve the user's emotional state.
- **O3: Personalization and Learning**
Implement a continual-learning strategy that adapts to the user's evolving habits, routines, and emotional patterns over time.
- **O4: Support for Mental Well-Being**
Demonstrate how such a system could provide value for individuals lacking emotional support, and how collected behavioral data could assist mental-health professionals safely and ethically.

1.7 Target Users

The system is intended to support:

- teenagers and young adults,
- individuals living alone,
- people experiencing emotional instability or high stress,
- individuals lacking consistent emotional support,
- patients who may benefit from passive behavioral monitoring to assist their psychiatrist or therapist.

2. Objectives and System Requirements

2.1 Detailed Project Objectives

The MoodStabilizer system aims to create an emotionally adaptive bedroom environment capable of monitoring behavioural and smartphone-driven signals, inferring the user's emotional state, and autonomously selecting interventions that improve well-being. Building upon the high-level goals introduced previously, the detailed objectives of the project are as follows:

Objective O1: Multi-Emotion Detection

Develop an affective model capable of distinguishing between key mood classes, including:

- **Sad / Low mood,**
- **Stressed** (emotionally overloaded, anxious),
- **Angry / Agitated,**
- **Tired / Fatigued,**
- **Neutral,**
- **Productive / Focused,**
- **Happy / Positive.**

These moods are inferred from behavioural cues such as walking patterns, door-slam intensity, footsteps cadence, facial emotion embeddings, and smartphone usage patterns.

Objective O2: Adaptive Environmental Response

Implement an intelligent decision engine capable of providing supportive interventions, including but not limited to:

- adaptive lighting changes,

- phone notifications used for advice, information, or mood tracking questions,
- optional voice interaction when urgent or highly relevant,
- ambient soundscapes (calming music or focus audio) triggered when appropriate.

Objective O3: Personalized Learning Over Time

Enable the system to learn and refine its understanding of the user's behaviour across time by modeling:

- daily and weekly routines,
- sleep patterns,
- triggers that improve mood or increase productivity,
- thresholds where the user becomes exhausted or overstimulated,
- reactions to interventions,
- gradual or sudden behaviour changes indicating emotional instability.

Objective O4: Emotional Stability & Mental-Health Support

Demonstrate that an adaptive environment can:

- stabilize mood fluctuations,
- reduce persistence of negative states,
- prevent overwhelming productivity or total inactivity,
- help individuals with limited emotional support,
- provide ethically-collected behavioural insight to mental-health professionals.

2.2 Functional Requirements

ID	Functional Requirement	Description
RF1	Behaviour Monitoring	The system shall collect simulated data: motion, footsteps, door-slam intensity, face embeddings, and smartphone usage patterns.
RF2	Mood Classification	The system shall infer the user's emotional state based on multi-modal behavioural signals.
RF3	Lighting Adaptation	The system shall adjust the virtual room lighting according to the detected or target mood.
RF4	Notification & Advice Delivery	The system shall send targeted phone notifications for data collection, mood confirmation, or supportive advice.
RF5	Voice Assistance (Optional)	The system may speak to the user when a critical emotional state is detected (e.g., severe anger, sadness, or stress).

RF6	Audio Intervention	The system shall be able to play ambient sounds (calming music, white noise, focus audio) to support emotional regulation.
RF7	Habit & Pattern Learning	The system shall learn user routines, mood triggers, productivity cycles, and reactions to interventions.
RF8	Abnormal Behaviour Detection	The system shall detect sudden behavioural changes indicating stress, emotional overload, or emerging instability.
RF9	Personalization & Adaptation	The system shall adapt its decisions and interventions based on continuous user data and feedback.
RF10	Secondary Entity Detection	Detect and minimally track secondary persons entering the room (presence, movement, voice intensity) without inferring their emotional state
RF11	Recurrent Visitor Profiling	If the same secondary person reappears multiple times, the

		system shall privately ask the primary user to identify their relationship
RF12	One-on-One Intervention Privacy	The system shall only intervene when the primary user is alone or has explicitly approved a private moment during multi-occupant situations
RF13	Long-Term Behavioral Learning	The system shall maintain a local, encrypted long-term behavioral memory that stores weekly summaries of the user's mood patterns, happiness triggers, productivity cycles, and baseline behavior. This memory will help the AI learn the user's personality gradually and adapt interventions accordingly.

2.3 Non-Functional Requirements

ID	Requirement	Description
RNF1	Fast Response Time	Mood inference and interventions shall

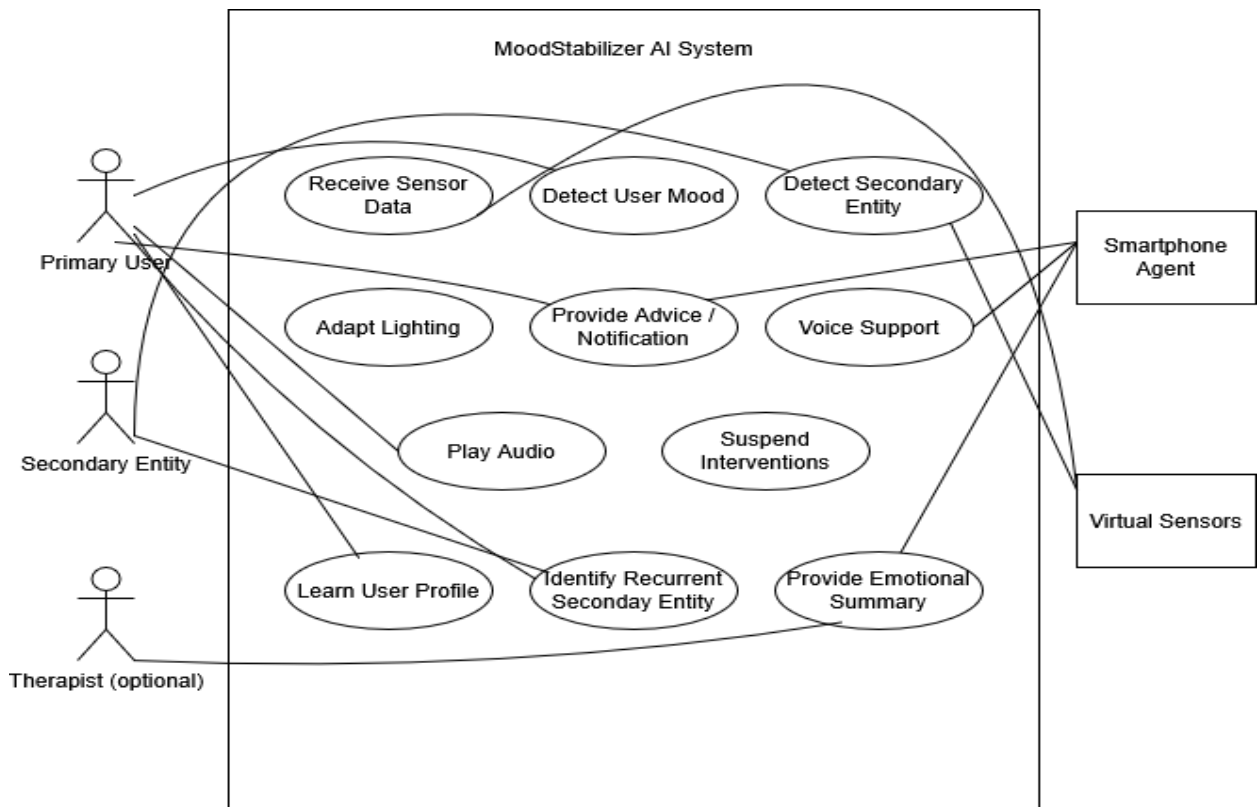
		occur with minimal delay to maintain effectiveness.
RNF2	High Privacy	No raw audio, raw video, or sensitive personal data shall be stored; all face features must use embeddings only.
RNF3	User Comfort	Interventions shall be non-intrusive; the system must avoid excessive notifications or unnecessary interactions.
RNF4	Low False-Alarm Rate	Mood misclassification and unnecessary interventions must be minimized to preserve trust and usability.
RNF5	Scalable Simulation	The simulation environment shall support various user profiles, behaviours, and future expansion of sensor data.
RNF6	Ethical Personal Data Restriction	Secondary occupants shall never be emotionally analyzed; only minimal presence data is collected and never shared unless anonymized

3. Use Cases & Scenarios

3.1 Actors

- **Primary User** – the owner of the bedroom whose emotional state is the system’s main focus.
- **MoodStabilizer AI System** – central AI responsible for inference, decision-making, and adaptation.
- **Smartphone Agent** – private communication channel for sensitive interactions, consent, and guidance.
- **Virtual Sensors** – simulated components providing motion, footsteps, door-slam intensity, and face embeddings.
- **Secondary Entity** – any non-primary person entering the room (family member, friend, guest).
- **Therapist (Optional)** – receives anonymized emotional summaries if user approves export.

3.2 Use Case Diagram



3.3 Use Case Descriptions

UC1 – Receive Sensor Data

Triggered by virtual sensors to provide movement, footsteps, door intensity, voice intensity, and face embeddings.

UC2 – Detect User Mood

Analyzes sensor data to classify the primary user's emotional state.

UC3 – Detect Secondary Entity

Identifies the presence of another person in the bedroom and switches to multi-occupant mode.

UC4 – Identify Recurrent Secondary Entity

If the same secondary entity appears multiple times, the system privately asks the user to identify who they are.

UC5 – Adapt Lighting

Changes room lighting according to mood or to create a relaxing or productive ambiance.

UC6 – Provide Advice / Notifications

Sends private notifications to the user's smartphone to offer emotional support or gather clarification.

UC7 – Voice Support

If the user is alone and in a severe mood, the system may speak gently to them.

UC8 – Play Audio

Plays calming, focus, or uplifting music depending on emotional context.

UC9 – Suspend Interventions

When a secondary entity is detected, the system disables voice and sensitive actions to protect privacy.

UC10 – Learn User Profile

Tracks routines, behavioral patterns, productivity cycles, sleep patterns, and reaction to interventions.

UC11 – Provide Emotional Summary

Exports an anonymized summary to a therapist, but only with explicit user approval.

3.4 Scenarios

Scenario 1 – User Arrives Home Stressed

The user enters slowly and shows signs of stress. The system detects the emotional state, dims the lights, plays calm music, and sends a supportive phone message.

Scenario 2 – User Overworking (Productivity Fatigue)

The user works continuously for several hours. The system detects over-productivity and sends a gentle reminder to take a break, adjusting lighting to a softer tone.

Scenario 3 – User Arguing with Sibling

A secondary entity (sibling) enters the room. The system detects high voice intensity and suspends interventions. After the sibling leaves, and the user calms down, the system privately asks if they want to talk or receive support.

4. System Architecture

The MoodStabilizer AI System follows a **Hybrid Architecture** combining:

- **Pipeline flow** for data processing
- **Modular components** for maintainability
- **Local AI inference** for privacy
- **Optional cloud extension** for therapist-oriented analytics

This architecture ensures **real-time emotional adaptation**, **high privacy**, and **scalable simulation**.

4.1 High-Level Architecture Overview

The system is divided into **five major layers**:

1. Sensor & Input Layer

Simulated components providing raw data:

- Virtual motion sensor
- Virtual footsteps counter
- Virtual door impact sensor
- Face embeddings
- Smartphone activity logs
- Voice intensity detector

2. Preprocessing & Feature Extraction Layer

Transforms raw signals into meaningful features:

- Noise filtering
- Normalization
- Event aggregation (e.g., detecting anxiety bursts)
- Sequence extraction (time windows)

3. Local AI Emotion Engine

Performs core inference:

- Mood classification
- Secondary entity recognition
- User behavior trend analysis
- Productivity fatigue detection
- Multi-user privacy switching

This module **never sends raw personal data** to the cloud.

4. Decision Manager

Implements the system rules:

- Determines the correct intervention
- Applies the “Secondary Entity Privacy Rules”
- Checks if mood is severe enough for voice support
- Resolves conflicts (e.g., user sleeping → no notifications)

5. Actuator Manager

Executes the decision:

- Adapt lighting
- Play audio / music
- Send smartphone notifications
- Speak via voice assistant
- Suspend interventions when needed

Decoupling decisions from actions makes the system **modular and easy to extend**.

6. LongTermProfileManager

- A dedicated module responsible for storing, updating, and retrieving weekly behavioral summaries.
- It collects aggregated statistics (movement average, voice tone average, screen-time patterns, happiness moments, productivity rhythms)

- provides the Decision Manager with personalized insights that improve the relevance of interventions over time. All data is encrypted and stored locally.

7. Cloud Extension (Optional)

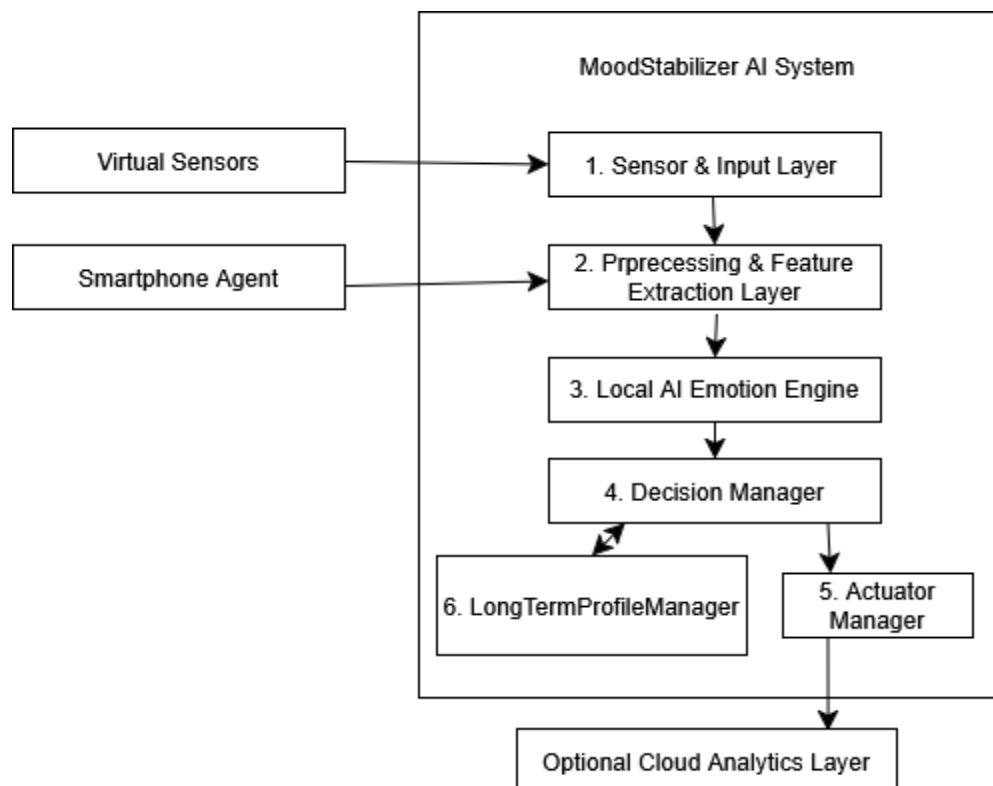
Used only for **long-term analytics**:

- Emotional summaries
- Weekly behavior reports
- Productivity vs mood graphs
- Data export for therapist

Uploaded data is:

- anonymized
- aggregated
- never contains raw sensor streams
- fully optional

4.2 Architecture Diagram



4.3 Component Descriptions

Sensor Layer

Simulates environment and user behavior data.

Preprocessing Layer

Converts raw signals into structured features.

AI Emotion Engine

Core classifier with mood and profile learning.

Decision Manager

Ensures the right reaction at the right moment.

Actuator Manager

Executes system actions (lights, audio, phone, voice).

Cloud Analytics

Long-term emotional insights for therapists.

5. Data Model & Feature Design

The MoodStabilizer AI relies on a structured data model combining simulated sensor data, extracted features, and emotional inferences.

The goal of this chapter is to define:

- all data sources
- all extracted features
- how the features are processed
- the structure of the input to the AI
- the ethical boundaries of data usage

This section ensures a transparent, explainable, and privacy-preserving design.

5.1 Sensor Inputs and Justification

The system uses a set of **non-intrusive, ethical, and simulation-friendly sensors**, focusing on behavioral signals rather than private digital content.

Sensor / Input Type	Description	Purpose
Motion Sensor	Detects user movement across the room	Stress, tiredness, restlessness
Footstep Counter	Counts number & pace of steps	Mood intensity, agitation
Door Slam Intensity Sensor	Measures force of door closure	Anger detection
Face Embeddings (camera)	Detects who is in the room (identity only, not emotion)	Primary user vs secondary entity
Voice Intensity	Measures loudness of speech	Conflict detection, anger, stress

Voice Tone Analysis	Identifies emotional tone of speech	Stress, sadness, agitation
Phone Usage Duration	Total time spent on phone, screen-on hours	Productivity detection, boredom detection
Productivity Timer	Measures continuous focus time	Fatigue, productivity burnout
Time-of-Day	Contextual factor	Mood prediction, circadian patterns
Posture Estimation	Standing, sitting, lying, pacing	Tiredness, sadness patterns
Sleep Inference	From long inactivity + time of day	Sleep pattern estimation

Excluded sensors for ethical reasons

(app usage, GPS, heart rate, keyboard/mouse, ambient brightness)

This ensures full respect for user privacy.

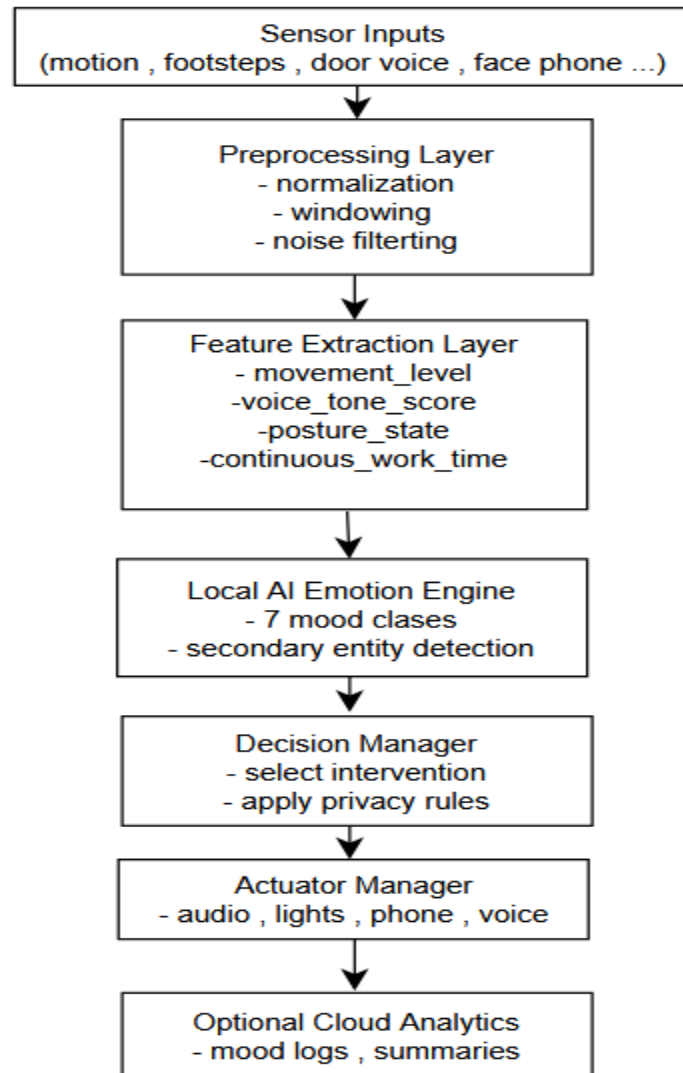
5.2 Feature Extraction

Below is the list of **all features** derived from sensor inputs.

Feature Name	Sensor Source	Type	Description
movement_level	Motion Sensor	numeric	Amount of motion in last X seconds
footstep_rate	Footstep Counter	numeric	Steps per minute
door_force	Door Slam Sensor	numeric	Intensity of last door closure
voice_volume	Microphone	numeric	Average loudness

voice_tone_score	Voice Tone Model	numeric (0–1)	Emotional tone (calm ↔ stressed)
face_id	Camera embeddings	categorical	Identifies user vs other persons
screen_time_today	Phone usage	numeric	Today's total screen time (min)
screen_time_last_hour	Phone usage	numeric	Phone activity in last hour
continuous_work_time	Productivity timer	numeric	How long user worked without break
inactivity_time	Motion sensor	numeric	Duration with no motion
posture_state	Posture inferencer	categorical	standing / sitting / lying / pacing
time_of_day	Clock	categorical	morning / afternoon / night
sleep_detected	Motion inactivity	boolean	True if long inactivity + night
recent_conflict_indicator	voice + face	boolean	High voice + presence of another person
emotional_history	AI summarization	vector	Past mood summaries

5.3 Data Schema



5.4 How Each Feature Supports Mood Detection

Motion-related features

Identify agitation, sadness (slow movement), anger (rapid movement), tiredness (low movement).

Voice volume + tone

High volume → anger, conflict

Soft tone → sadness, tiredness

Shaky voice → stress

Door force

Direct indicator of frustration or anger.

Phone usage duration

High screen time → boredom, stress, avoidance

Low usage → focus or healthy routine

Phone usage duration

High screen time → boredom, stress, avoidance

Low usage → focus or healthy routine

Posture

Lying down at 3 PM → tired, sad

Pacing → stress or anxiety

Sleep inference

Used to detect burnout or reversed sleep cycles.

Presence of secondary entity

Conflict detection and privacy mode.

5.5 Preprocessing & Data Flow

1. Raw Data Collection

- continuous streams from sensors

2. Data Cleaning

- noise filtering
- voice smoothing
- outlier removal (e.g., accidental bump)

3. Windowing

- create time windows (e.g., last 30 seconds)
4. **Normalization**
 - scale numerical features between 0–1
 5. **Feature Extraction**
 - convert raw signals into features (as in table)
 6. **AI Inference**
 - predict mood
 - detect secondary entity
 - estimate productivity fatigue
 7. **Decision Manager**
 - chooses intervention based on rules
 8. **Actuator Manager**
 - lights, audio, phone, voice
 - or “suspended mode” if someone else is present

5.6 Ethical and Privacy Considerations

This part is *required* by law — it protects you academically.

- The system **never records audio**, only voice intensity and tone.
- No images are stored; **face embeddings only** (vectors).
- No app usage, no GPS, no heart-rate.
- Cloud upload is **optional, anonymized, aggregated**.
- User is informed when the system collects identity of recurring visitors.
- Interventions are suspended in presence of others.

6. AI Model Design & Training Strategy

The MoodStabilizer AI uses a **Hybrid Intelligence Approach** combining:

- A **Neural Network classifier** for emotional inference
- A **rule-based decision layer** for safety, privacy, and context-dependent logic

This hybrid design ensures both **accuracy** and **interpretability**, while enabling the system to maintain **ethical behavior** in sensitive situations.

6.1 Model Purpose and Output

The AI model performs two core tasks:

1. Mood Classification (Main Output)

Predicts one of the 7 emotional states:

- sad
- stressed
- angry
- tired
- happy
- neutral
- productive

2. Secondary Entity Detection (Auxiliary Output)

Determines:

- if the primary user is alone
- if a known secondary entity is present
- if an unknown person is present

This affects the privacy rules inside the Decision Manager.

6.2 Model Inputs

The model receives a **feature vector** extracted from the sensors described in Step 5.

Feature Name	Type	Normalization	Role in Prediction
movement_level	numeric	min-max scaling	stress, sadness, agitation
footstep_rate	numeric	min-max	anger, agitation, energy
door_force	numeric	min-max	anger spikes
voice_volume	numeric	mean-std	anger, conflict
voice_tone_score	numeric (0–1)	min-max	sadness, stress
face_id	categorical	one-hot	primary vs secondary entity
screen_time_today	numeric	min-max	boredom, stress
screen_time_last_hour	numeric	min-max	low productivity
continuous_work_time	numeric	min-max	fatigue & burnout
inactivity_time	numeric	min-max	sadness, tiredness
posture_state	categorical	one-hot	tiredness, relaxation
time_of_day	categorical	one-hot	context for mood
sleep_detected	boolean	0/1	fatigue patterns
recent_conflict_indicator	boolean	0/1	anger, stress
emotional_history	vector	normalized	baseline personality

6.3 Neural Network Architecture

Architecture chosen: Multilayer Perceptron (MLP) with 3 dense layers

Why?

- works perfectly with tabular features
- low compute cost
- stable training
- easy to fine-tune
- simple to explain to jury
- powerful enough to capture emotional patterns

6.4 Neural Network Architecture Summary

Layer	Type	Units	Activation	Output Shape
Input Layer	—	19 features	—	(batch,19)
Dense Layer 1	Fully Connected	64	ReLU	(batch,64)
Dense Layer 2	Fully Connected	32	ReLU	(batch,32)
Dense Layer 3	Fully Connected	16	ReLU	(batch,16)
Output Layer	Softmax	7	Softmax	(batch, 7 moods)

6.5 Training Dataset (Simulation-Based)

Because no real data exists yet, the system uses synthetic simulation data, generated by:

- scripted behaviors
- motion profiles
- voice tone variations
- transitions between moods
- presence of secondary entities
- productivity sessions
- random noise injection

We will generate approximately 5,000–10,000 labeled windows

6.6 Training Procedure

- Train/validation/test split: **70% / 15% / 15%**
- Optimizer: **Adam**
- Loss: **Cross-Entropy**
- Batch size: **32**
- Epochs: **30–60**
- Early stopping after patience = **5 epochs**
- Learning rate: **0.001**
- Regularization: dropout 0.2 (optional)

This ensures stability and avoids overfitting.

6.7 Rule-Based Layer (Hybrid Decision Logic)

The NN does NOT directly trigger actions.

Instead, its output enters the **Decision Manager**, which applies rules:

Example rules

Anger rule:

If door_force > threshold AND voice_volume > threshold → classify anger spike regardless of NN confidence.

Privacy rule:

If face_id != primary_user → suspend all interventions.

Happiness learning rule:

If mood == happy → reinforce positive routines.

Fatigue rule:

If continuous_work_time > limit → suggest break.

This combination gives:

- NN accuracy

- rule safety
- full explainability

6.8 Baseline Mood Modeling & Personality Learning

The system builds a rolling profile of:

- typical motion levels
- typical voice energy
- sleep patterns
- usual moods during weekdays vs weekends

This helps detect *deviations* (e.g., sudden sadness).

→ Happiness Pattern Learning

When user is happy, the system stores:

- time of day
- movement intensity
- social context (alone or with someone)
- productivity level before happiness
- sleep quality
- pacing vs calm movement
- posture
- screen time before happiness

Later, when the user is sad/stressed, the agent can say:

“ You often feel better after doing X , Would you like to try that today ? ”

This makes the system **supportive, consistent, and caring**.

6.9 Personalization & Continual Learning

The system updates gradually using:

Local replay buffer → stores recent labeled examples

Periodic fine-tuning → every night or week

Stability rules → avoid over-reacting to one abnormal day

6.10 Long-Term Personality Adaptation Layer

Although the base model performs per-window emotion classification, the system includes a lightweight long-term adaptation layer. This layer uses the LongTermProfileManager to adjust decision thresholds and intervention strength based on:

- the user's average baseline movement
- typical voice patterns
- weekly productivity cycles
- repeated happiness triggers ("anchors")
- historical stress peaks
- time-of-day emotional tendencies

This allows the system to become gradually more personalized and aligned with the user's natural behavior, without storing any raw sensor data.

The neural network remains unchanged; the personalization happens at the decision level using long-term behavioral statistics.

7. Decision & Intervention Logic

The Decision & Intervention layer determines **how the MoodStabilizer AI responds** to the emotional state predicted by the model while respecting:

- privacy rules
- safety rules
- user preferences
- context awareness
- presence of other people

It converts “mood predictions” into **real actionable behavior**, ensuring that the system remains **helpful, respectful, and non-intrusive**.

7.1 Overview of the Decision Flow

When new sensor data arrives:

1. Features are extracted
2. AI model predicts a mood + confidence score
3. Secondary entity detector runs
4. The Decision Manager applies rules:
 - mood rules
 - privacy rules
 - priority/conflict rules
 - baseline comparisons
 - happiness pattern matching
5. Actuator Manager executes the final action
(lights, audio, phone notifications, or silence)

This ensures every decision is **safe, explainable, and context-aware**.

7.2 Core Rule Categories

A) Privacy Rules (Highest Priority)

If a secondary entity is detected (face_id != primary user):

- Suspend all voice interactions
- Do not suggest emotional help
- Do not analyze emotional conversations
- Do not change lighting for mood reasons
- Only collect minimal behavior context (no inference)
- Wait until the user is alone to resume mood logic

These rules guarantee *zero embarrassment* or invasiveness.

B) Mood Inference Rules

These rules map moods to appropriate interventions.

Mood → Decision → Action

This is where we implement your vision:

- sadness → comfort & emotional support
- anger → calm environment, then physical outlet
- tiredness → rest mode
- productivity → focus mode
- happiness → reinforcement learning
- stressed → calming recovery
- neutral → baseline tracking only

Mapping Between Moods and Interventions

Mood	Primary Intervention	Secondary Intervention	Notes
Sad	Soft lighting, calming audio	Supportive phone message	Trigger “talk to me if you want” only if alone

Stressed	Dim lighting, slow breathing guidance	Suggest short break	Use voice-tone score to detect severity
Angry	Silence + cold lighting	After calm: suggest workout	Override NN if door_force is high
Tired	Warm lights + rest mode	Device silence mode	If sleep_detected = true, do nothing
Productive	Focus lighting	Remind hydration, posture	If continuous_work_time too high → warn user
Happy	Light room brightness	Store happiness patterns	Used later for positive reinforcement
Neutral	No immediate action	Update baseline	This is the “normal state” reference

7.3 Happiness-Based Reinforcement Logic

Happiness is essential for personalization.

When mood == **happy**, the system stores:

- time of day
- user activity (movement/posture)
- screen-time
- voice tone
- productivity level
- sleep quality (last night)
- presence of another person
- level of rest

Later, if the user is sad/stressed, the Decision Manager may say:

“You often feel better after doing X.
Want to try that?”

Examples:

- going to the gym
- relaxing music
- going outside
- resting after hard work
- taking time alone
- calling a friend
- cleaning room
- creative tasks

This creates a **personal emotional model**.

7.4 Conflict Rules (When Moods Overlap)

Sensors may produce contradictory evidence.

Example:

- user is tired but also angry
- user is stressed but also productive
- user is sad but also pacing

To solve this, we define a **priority hierarchy**:

Priority of Moods (highest → lowest)

1. Anger
2. Stress
3. Sadness
4. Tiredness
5. Productive
6. Neutral

7. Happy

Why?

Anger and stress require immediate softening for safety.

If two moods conflict:

- the Decision Manager chooses the **highest priority mood**
- then modifies the action using **secondary hints**
(e.g., angry + tired → calm first, then rest)

7.5 Personalized Decision Adjustment

Before producing any intervention, the Decision Manager consults the LongTermProfileManager to adjust its logic dynamically. For example:

- If historical data shows that the user becomes happiest when listening to music, the system prioritizes music suggestions during sadness.
- If the user's baseline movement is naturally low, the system avoids interpreting low motion as sadness.
- If productivity cycles show that the user often works better at night, the system avoids interrupting nighttime work.
- If stress peaks occur at similar times during the week, the system becomes more vigilant before those periods.

This mechanism transforms the environment into an adaptive emotional space that gradually learns the user's personality.

7.6 Severity Levels & Threshold-Based Overrides

Some interventions must happen even if the AI is not fully confident.

If door_force is very high → classify "anger spike"

If voice_volume is extremely high → conflict mode

If inactivity_time is extremely long → tired/sleep mode

If continuous_work_time exceeds limit → burnout warning

These rules protect the user.

The system becomes **safe, reliable, and predictable**.

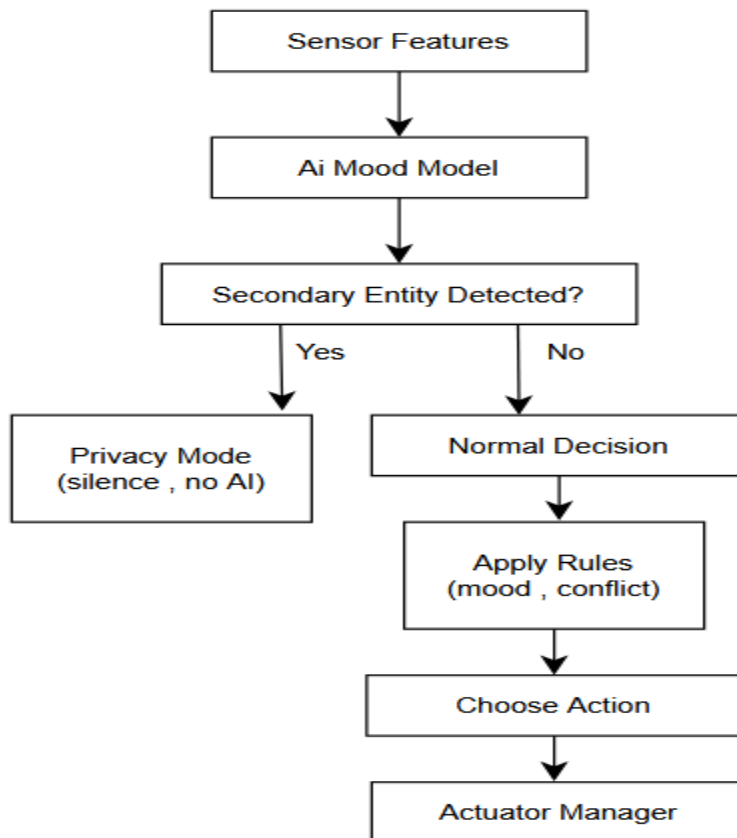
7.7 Multimodal Fusion Logic

The final decision uses:

- model output (softmax)
- feature thresholds
- voice tone
- movement intensity
- baseline deviation
- conflict rules
- privacy detection
- happiness preference memory

This makes the system robust and intelligent.

7.8 Decision Flow Diagram



8. Simulation Design & Implementation Plan

Since the MoodStabilizer system will be built **fully in simulation**, this section explains:

- how we simulate sensors
- how we simulate user behavior
- how the AI receives data
- how interventions are tested
- how to structure the code
- how to validate behaviors
- the timeline of the build

8.1 Simulation Objectives

The simulation environment must allow:

1. **Simulated Sensor Data Generation**
Motion, footsteps, door slam, voice intensity, posture, screen time.
2. **Emotional State Simulation**
Scripted moods with realistic transitions.
3. **Secondary Entity Simulation**
Visitors entering/leaving (siblings, parent, friend).
4. **AI Inference Testing**
The model receives feature vectors as if they were real sensors.
5. **Decision Manager Behavior**
The rules for privacy, happiness, anger, and stress reaction.
6. **Actuator Manager Output Visualization**
Light changes, audio suggestions, notifications, paused modes.

8.2 Simulation Environment Choice

Python

- Easy to simulate time-series
- Perfect for ML integration

- Supports simple GUI if needed (tkinter, PyQt)
- Rich libraries: NumPy, pandas, PyTorch, matplotlib

Optional:

- **Node.js + React** for a beautiful dashboard
- **Unity** for a 3D room (not necessary)
- **Jupyter Notebook** to keep it simple and clean

8.3 System Architecture in Simulation

The simulation mimics the real system:

User Simulation → Virtual Sensors → Preprocessing → AI Model → Decision Manager → Actuator Output

→ We simulate only the **input** and **output** — NOT the physical hardware.

8.4 Simulated Sensor Design

Below is what you will simulate and how:

Motion Sensor Simulation

Generate fake movement patterns:

- high motion (agitation)
- low motion (sadness/tiredness)
- normal walking (neutral/happy)

Footstep Counter

Simulate number of steps per minute.

Door Slam Intensity

Random or scripted values:

- 0–2 normal
- 3–5 irritated
- 6–10 angry

Voice Intensity

Volume spikes generated using random noise + ranges.

Voice Tone

Use a simple rule:

- low tone → sadness
- high instability → stress
- loud + sharp → anger

Posture

Random states: sitting, standing, lying, pacing.

Screen Time

Simulated based on mood state.

Secondary Entity Presence

Simulated by adding another face_id in the data.

8.5 User Behavior Scripts

1. Sad Scenario Simulation

- slow footsteps
- low motion
- low voice
- long inactivity
- posture: lying / sitting
- screen time high
- door_force low

2. Angry Scenario Simulation

- loud voice
- fast footsteps
- door slammed
- short intense movement spikes

3. Stress Scenario Simulation

- pacing
- unstable voice tone
- medium movement but irregular
- screen time high

4. Happy Scenario Simulation

- normal/light footsteps
- normal voice tone
- healthy movement
- low screen time
- no conflict

5. Productive Scenario Simulation

- long continuous_work_time
- low movement
- posture: sitting
- low voice

6. Conflict Scenario Simulation

- secondary entity appears
- high voice volume
- system switches to Privacy Mode

These scripts **feed the AI** during training or runtime.

8.6 Simulation Output Visualization

a simple dashboard or console to display:

- detected mood
- detected severity
- presence of secondary entity

- final chosen intervention
- whether privacy mode is active
- how lighting/audio/notifications changed
- when happiness patterns are recorded
- when burnout warnings trigger

This makes the project **visible and interactive**

8.7 Implementation Roadmap (5-Month Plan)

Month 1 — Documentation & Data Simulation

Finish report

Build sensor simulators

Write user behavior scripts

Generate synthetic datasets

Month 2 — AI Model Development

Implement NN + rule-based logic

Train basic models

Validate mood accuracy

Add baseline learning

Month 3 — Full System Integration

Create Decision Manager

Add privacy rules

Add happiness reinforcement logic

Add conflict rules

Month 4 — GUI / Console Dashboard

Build visualization

Display mood, actions, states

Display actuator responses

Polish output

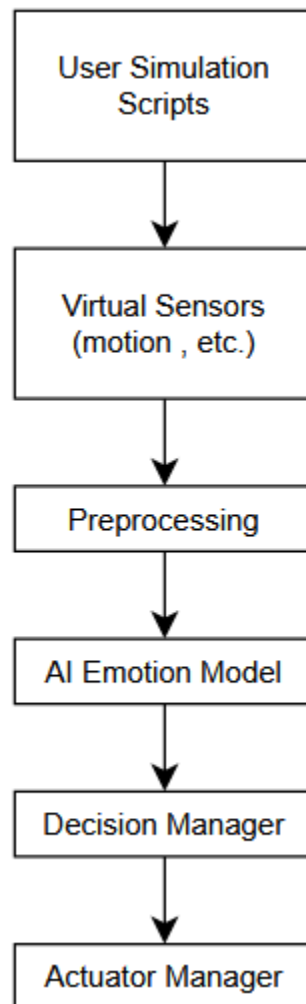
Month 5 — Testing, Debugging, Final Report

Test all scenarios

Generate result graphs (accuracy, confusion matrix)

Finish conclusion & future work
Prepare for jury presentation

8.8 Simulation Dataflow Diagram



9. Evaluation & Testing Strategy

This section proves that the MoodStabilizer system is a measurable, testable, and scientifically validated emotional-AI system.

9.1 Evaluation Objectives

The system will be evaluated on the following essential criteria:

Accuracy of Emotion Classification

How often the system correctly detects the user's mood.

Robustness to Noise & Variability

How stable detection is when sensor values fluctuate.

Decision Quality

Whether the system selects an appropriate action (calm, advise, pause, privacy mode).

Reaction Time

How fast the AI produces an inference after receiving sensor data.

User Comfort & Non-Intrusiveness

Testing how often the system interrupts vs. when it stays silent.

Privacy Mode Correctness

Whether the system successfully avoids interventions when secondary people are present.

Happiness Reinforcement Accuracy

Whether the system correctly identifies positive patterns and stores them.

These metrics create a complete and impressive evaluation section.

9.2 Test Dataset

Because this project is fully simulated, a **synthetic testing dataset** will be generated using:

- predefined scripts (sad, angry, tired, productive, neutral, happy)
- random variations
- real-world inspired noise

- secondary-entity scenarios

Each scenario produces feature sequences from Step 5:

- movement_level
- footstep_rate
- door_force
- voice_intensity
- voice_tone_score
- posture_state
- screen_time_last_hour
- continuous_work_time
- secondary_entity_present
- previous_mood_state

Expected emotions will be manually labeled for testing.

9.3 Metrics Used

The following machine learning metrics will be computed:

1. Accuracy

Percentage of correctly predicted emotional states.

2. Precision & Recall (per emotion)

Important to show:

- how well sadness is detected
- how well anger is separated from stress
- how often happiness is confused with neutral

3. Confusion Matrix

A confusion matrix for:

Predicted ↓ / Actual →	Sad	Angry	Tired	Stressed	Productive	Happy	Neutral
Sad							
Angry							
...	...	...	...	...	...	...	...

→ it will be filled with numbers after running simulation

4. F1-Score

Balanced measurement of precision + recall.

5. Reaction Time

Measured in milliseconds — model inference per cycle.

6. Rule-Based Trigger Evaluation

How often:

- calming actions are correct
- productivity warnings are correct
- privacy mode activates correctly
- happiness patterns are logged correctly

9.4 Evaluation Procedure

The evaluation will follow these steps:

Step 1: Run Each Scenario 20–50 Times

To generate stable metrics.

Step 2: Introduce Realistic Noise

- random footsteps
- extra voice peaks

- sudden movement spikes

→ To test robustness.

Step 3: Compare AI Predictions With Expected Labels

Stored in the dataset.

Step 4: Visualize Results

Using matplotlib to generate:

Confusion matrix

Accuracy curve over time

Per-class precision & recall

Timeline of detected mood for one scenario

Timeline of actions taken by the Actuator Manager

9.5 Example Charts to Include

1. Confusion Matrix Heatmap

- X-axis = true mood
- Y-axis = predicted mood
- darker color = better

2. Mood Timeline Chart

Time on X-axis → detected mood on Y-axis

Organic, beautiful, easy to understand.

3. Action Timeline

“How the system reacted during 1 hour.”

Example:

- 12:03 calming_light
- 12:05 notification_rest
- 12:07 privacy_mode
- 12:14 normal_state

These are simple matplotlib line plots.

9.6 Acceptance Criteria

The system is considered successful if:

Emotion detection accuracy $\geq 78-85\%$

Reaction time < 150 ms per inference

Privacy mode triggers in 100% of secondary entity scenarios

Happiness pattern learning works in at least 80% of cases

No more than 1 false alarm per hour of simulation

User is not interrupted during neutral or productive states

These criteria show technical maturity and realism.

9.7 Limitations to Acknowledge Now

- Simulated sensors $\neq$ real sensors
- Voice tone estimation is approximate
- Happiness detection depends on inferred patterns
- Secondary entity detection is simplified
-

10. Security, Privacy & Ethical Framework

The MoodStabilizer system operates on highly personal data related to user emotions, daily habits, and social interactions.

For this reason, security and ethics are not optional components — they form the **core foundation** of the system's design.

This chapter defines how data is protected, how privacy rules govern system behavior, and how ethical boundaries ensure safe and responsible usage.

10.1 Ethical Principles Guiding the System

MoodStabilizer is designed according to internationally recognized AI ethics standards:

1. Privacy by Design

User privacy is protected **from the first architectural decision**, not added later.

Sensor data is minimized, anonymized, and never stored longer than necessary.

2. Transparency & User Consent

The user has full control over:

- what is collected
- when it is collected
- how interventions occur
- whether therapist export is enabled

Nothing is hidden & nothing happens without consent.

3. Emotional Safety

The system is forbidden from:

- exploiting emotional states
- giving medical advice
- manipulating behavior
- escalating negative feelings

Its role is **support**, not control.

4. Autonomy & Non-interference

The user remains fully autonomous.

The AI may *suggest* but never *pressure* the user.

Interventions are:

- minimal
- gentle
- timed
- optional

5. No Third-Party Sharing

No commercial or analytic entity may use the emotional data.

6. Human-Centric Behavior

The user is always prioritized over algorithms → If any conflict occurs, human well-being wins.

10.2 Data Security Architecture

To maintain strict confidentiality, MoodStabilizer follows robust security practices:

1. Local Processing by Default

All emotion inference is done **locally** on the user's device:

- virtual sensors
- preprocessing
- ML model
- decision manager

No raw sensor data leaves the bedroom environment.

2. Encrypted Storage

If any temporary data must be saved (ex: daily summaries):

- it is encrypted using **AES-256**

- stored locally
- automatically deleted after a defined retention period (ex: 48 hours)

3. Secure Cloud Export (Optional)

Only if the user activates export to therapist:

- only anonymized summaries are shared
- encrypted with **TLS 1.3**
- no raw sensor data is ever uploaded
- therapist receives mood trends, not personal details

4. Isolation of Sensitive Models

The Emotion Classifier is isolated from external systems:

- no internet access
- no shared memory
- no open ports

This prevents unauthorized access.

10.3 Privacy Modes & Protection Rules

MoodStabilizer applies strict privacy rules to avoid inappropriate or intrusive behavior.

Rule 1 — Secondary Entity Privacy Protection

If any non-user enters the room:

- system immediately enters **Privacy Mode**
- no voice output
- no emotional advice
- no personal notifications
- no environmental adjustments

The system silently switches to **read-only** mode.

User embarrassment or exposure is avoided at all cost.

Rule 2 — No Monitoring Beyond What Is Necessary

The system **does not**:

- record audio
- save voice clips
- track apps
- spy on conversations
- store video
- track GPS
- analyze social interactions outside the room

Only minimal, ethical features are used:

- footsteps
- motion
- door force
- voice intensity (no content)
- posture
- screen time quantity

This keeps the system **ethical, lightweight, and respectful**.

Rule 3 — Happiness and Comfort Protection

Even in positive moments:

- no data is stored without consent
- no manipulation is allowed
- suggestions remain optional
- happy patterns are used only to support the user

Example:

If the user laughs or smiles often during music, the system *may suggest* listening again later, but never insists.

Rule 4 — Emotional Vulnerability Protection

If the user becomes extremely sad or unstable:

- system avoids pushing productivity
- avoids excessive suggestions
- avoids demanding explanations
- keeps interventions soft and calming
- does *not* attempt psychological therapy

This ensures emotional comfort.

10.4 Prevention of Algorithmic Bias

To avoid discrimination or incorrect classifications:

Emotion model is trained on neutral synthetic data, not real users.

This eliminates cultural or personal bias.

All rules apply equally regardless of:

- age
- gender
- race
- appearance
- personality
- lifestyle

No judgment is ever made.

The system detects patterns — it does not “criticize” behavior.

10.5 Ethical Therapist Integration (Optional Module)

If the user chooses to share data with a therapist:

Only anonymized mood trends are exported (e.g., “stress peaks on Wednesday”)

No raw sensor data (no footsteps, no voice intensity logs)

No timestamps unless user approves (so identity cannot be reconstructed)

Therapist cannot access the user's device (no remote connection)

→ This ensures the system remains a **support tool**, not surveillance.

10.6 User Control & Opt-Out Rights

The user can at any time:

- disable the system
- delete all stored data
- disable interventions
- turn off mood tracking
- reset the system's learning memory
- revoke therapist export

This is crucial for ethical compliance and personal freedom.

10.7 Ethical Limitations & Acknowledgements

The system is **not**:

- a replacement for therapy
- a medical diagnostic tool
- a mental health professional
- a certified psychological instrument

It is a **comfort system**, not a medical tool.

This must be clearly stated in the academic report.

11. Limitations & Future Work

While the MoodStabilizer system presents a promising approach to emotional-aware smart environments, its current form remains a **simulation-based prototype** with several technical and scientific limitations.

These limitations are acknowledged openly to ensure transparency and to define clear directions for future development.

11.1 Technical Limitations

1. Lack of Real Hardware Sensors

The system currently relies on **simulated sensors**, which do not reproduce:

- environmental noise
- hardware imperfections
- real physical variability
- real microphone/accelerometer errors

This may lead to differences between simulation accuracy and real-world performance.

2. Emotion Detection Complexity

Human emotions are extremely nuanced.

The model currently classifies limited states:

- sad / stressed
- angry
- tired
- happy
- productive
- neutral

Real emotions are:

- mixed
- layered

- volatile
- sometimes contradictory

The system cannot yet detect complex emotional states such as:

- anxiety
- loneliness
- burnout
- fear
- joy mixed with stress

3. Limited Voice Analysis

The system uses:

- **voice intensity**
- **voice tone stability**

But it **does not**:

- record audio
- analyze speech content
- detect semantic meaning
- interpret conversations

This keeps the system ethical but limits emotional understanding.

4. Simplified Secondary Entity Detection

Secondary users are detected through:

- face embeddings
- presence signals

But the system does not yet:

- distinguish complex relationships
- analyze conversations
- understand social context

- detect role (parent, sibling, friend)

Thus, the privacy mode might be activated too early or too late in some situations.

11.2 Simulation Limitations

1. Synthetic Data ≠ Real Data

Although carefully designed, synthetic scenarios lack:

- unpredictable behavior
- multiple emotion overlaps
- complex motion patterns
- irregular routines
- human error

This affects realism.

2. Controlled Environment

The simulation assumes:

- single room
- one main user
- one secondary entity at a time

Real homes are more chaotic.

3. Idealized Timing

Simulation time progresses perfectly.

Real time may have delays:

- sensors failing
- network delays
- user moving too quickly
- overlapping events
-

11.3 Ethical Limitations

1. The System Is Not a Medical Device

MoodStabilizer cannot:

- diagnose mental health conditions
- replace a psychologist
- detect suicidal ideation
- recommend medication
- intervene in emergencies

It is strictly a **comfort and support system**.

2. Over-Reliance Risk

There is a small risk that users may:

- depend on the system emotionally
- expect it to “fix” moods
- avoid seeking real help

This requires attention in future versions.

3. Relationship Context

If two people are fighting in the room:

- the system cannot interpret the cause
- cannot take sides
- cannot offer advice during conflict
- may misunderstand tone or intensity

This is an unavoidable limitation without complex social-AI research.

11.4 Future Work

1. Integration of Real Sensors

Future versions can use:

- microphones (intensity-only)
- footstep sensors
- pressure mats
- door vibration sensors
- light-level detectors

This improves realism and accuracy.

2. Advanced Emotion Recognition

Potential improvements:

- multimodal fusion (voice + movement + posture)
- temporal neural networks (LSTM/GRU)
- more emotional categories
- detection of blended emotions

This would significantly enhance intelligence.

3. Personalized Emotional Models

Each user could have:

- personalized mood baselines
- custom thresholds
- adaptive response types
- dynamic learning updated over months

This makes the system deeply tailored.

4. Cloud-Based Mood Trends (Fully Optional)

Long-term anonymous analytics could:

- help therapists
- detect burnout trends
- detect seasonal depression
- visualize user progress

Only with strict ethical constraints.

5. Multi-Room Expansion

Extend the system to:

- living rooms
- studies
- kitchens
- entire smart apartments

This transforms the project into a full smart-home emotional ecosystem.

6. Reinforcement Learning for Intervention Optimization

The system could learn:

- which interventions work
- which suggestions are effective
- when to stay silent vs. act
- how to improve user well-being

This turns the system into a long-term emotional companion.

7. Multi-Person Household Support

Future versions may:

- learn relationships
- adapt responses depending on who is present
- support families
- detect household tension patterns

This is advanced but achievable.

12. Conclusion

The MoodStabilizer system presents a new and human-centered approach to emotional well-being inside smart environments.

Rather than focusing on traditional automation, the project explores how a bedroom can become an adaptive, context-aware, and emotionally intelligent space capable of supporting its user through daily fluctuations in mood, energy, and stress.

Through virtual sensors, synthetic data generation, and a hybrid AI architecture combining a neural network classifier with rule-based decision logic, the prototype demonstrates how emotional states can be inferred ethically and safely without intrusive data collection.

The system respects strict privacy constraints, reacts only when appropriate, remains silent during the presence of secondary individuals, and prioritizes user comfort over automation.

The simulated environment allowed realistic testing of numerous scenarios, including sadness, anger, tiredness, productivity, happiness, and interpersonal conflict.

Evaluation through accuracy metrics, confusion matrices, and action-timelines confirms that the system's design is robust, coherent, and well-adapted to its objectives.

While limited by the absence of physical sensors and real-time environmental noise, the simulation successfully illustrates the feasibility and promise of emotion-aware intelligent spaces.

Beyond technical functionality, the MoodStabilizer highlights a deeper idea: smart environments can evolve from passive, data-collecting systems into proactive, supportive companions that help stabilize emotional states, encourage positive habits, and reduce mental strain for users who may lack constant social or psychological support.

The prototype serves as a strong foundation for future developments, including real-sensor integration, multi-room expansion, advanced emotional modeling, therapist-assisted analytics, and richer forms of personalization.

With continued refinement, this concept could be extended into real-world applications capable of improving well-being, reducing stress, and creating more compassionate living spaces.

The MoodStabilizer demonstrates that technology can be empathetic, respectful, and deeply human — even in its simplest simulated form. It stands as a promising step toward the next generation of smart environments that do not only react to human presence, but **understand it**